

PROYECTO PRÁCTICAS AULA 24

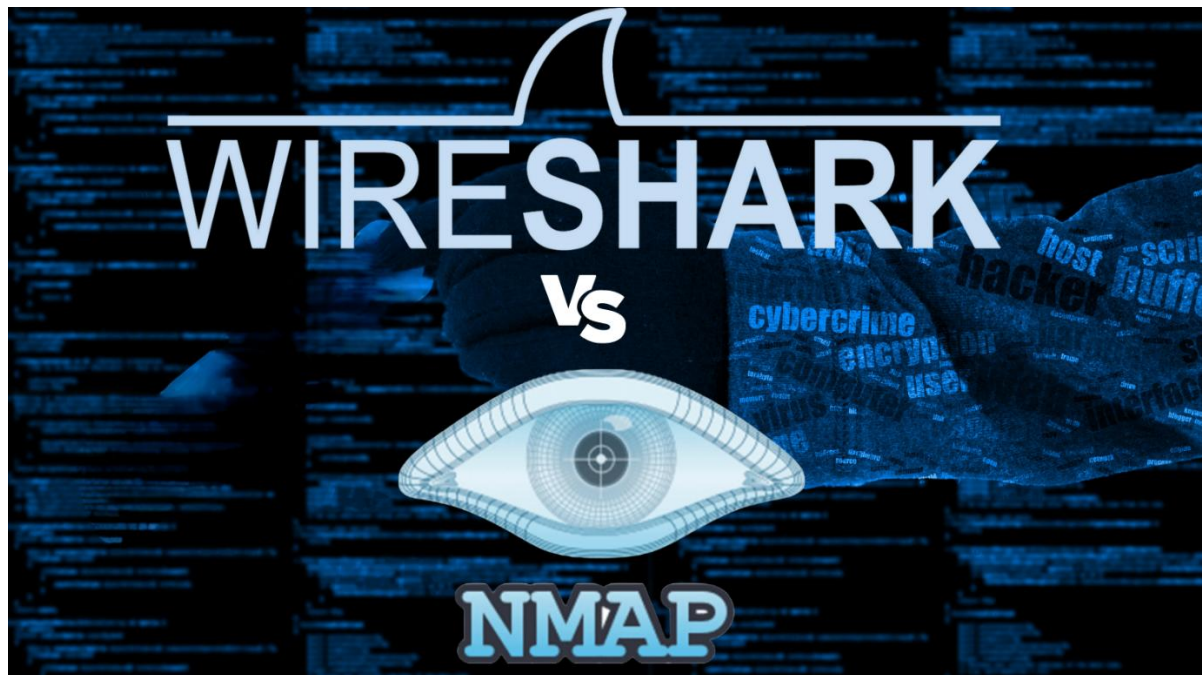
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Ciclo Formativo de Grado Superior en Administración de Sistemas
Informáticos en Red (ASIR) – Año 2025/2026.

WIRESHARK Y NMAP



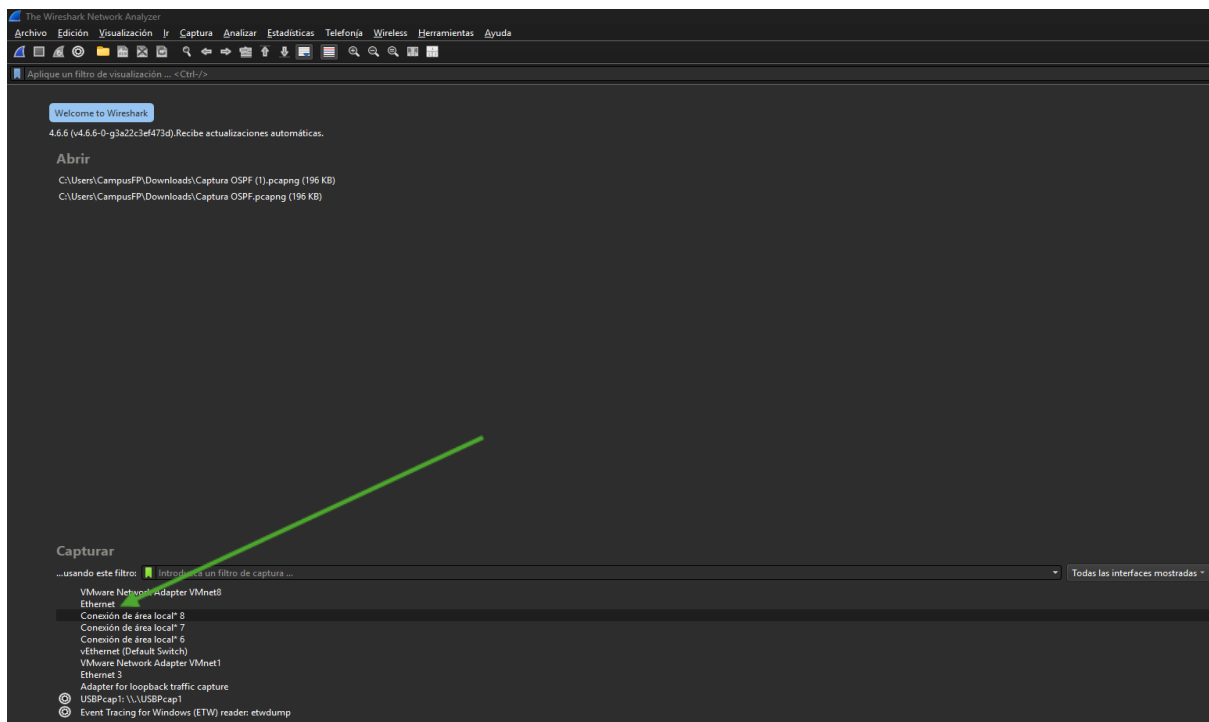
Índice:

WIRESHARK	3
NMAP	12
BIBLIOGRAFIA	24

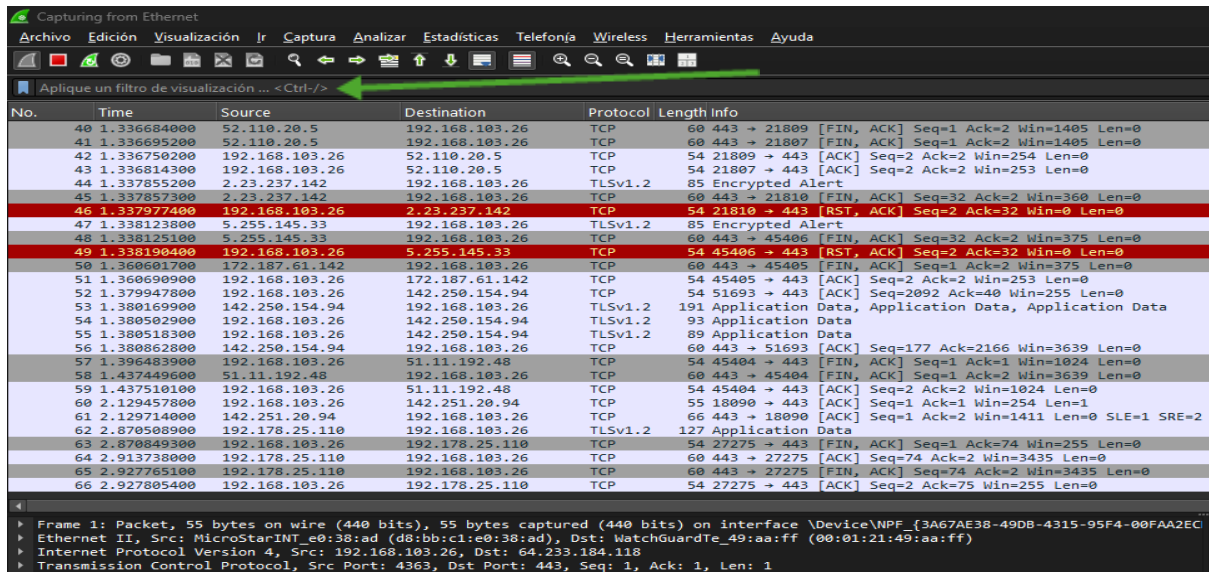
WIRESHARK

Wireshark es una herramienta de análisis de red que permite capturar y filtrar paquetes en tiempo real para estudiar el tráfico que circula por una red. Gracias a sus filtros, es posible identificar protocolos, direcciones IP, puertos, MAC y distintos tipos de comunicación, facilitando tareas de monitorización, diagnóstico y análisis de redes.

Primero en Wireshark elegimos lo que queremos capturar, en este caso Ethernet que es donde circula la mayoría de tráfico



Aquí en la barra de filtros es donde vamos a escribir para filtrar lo que queremos capturar



Applying a filter:

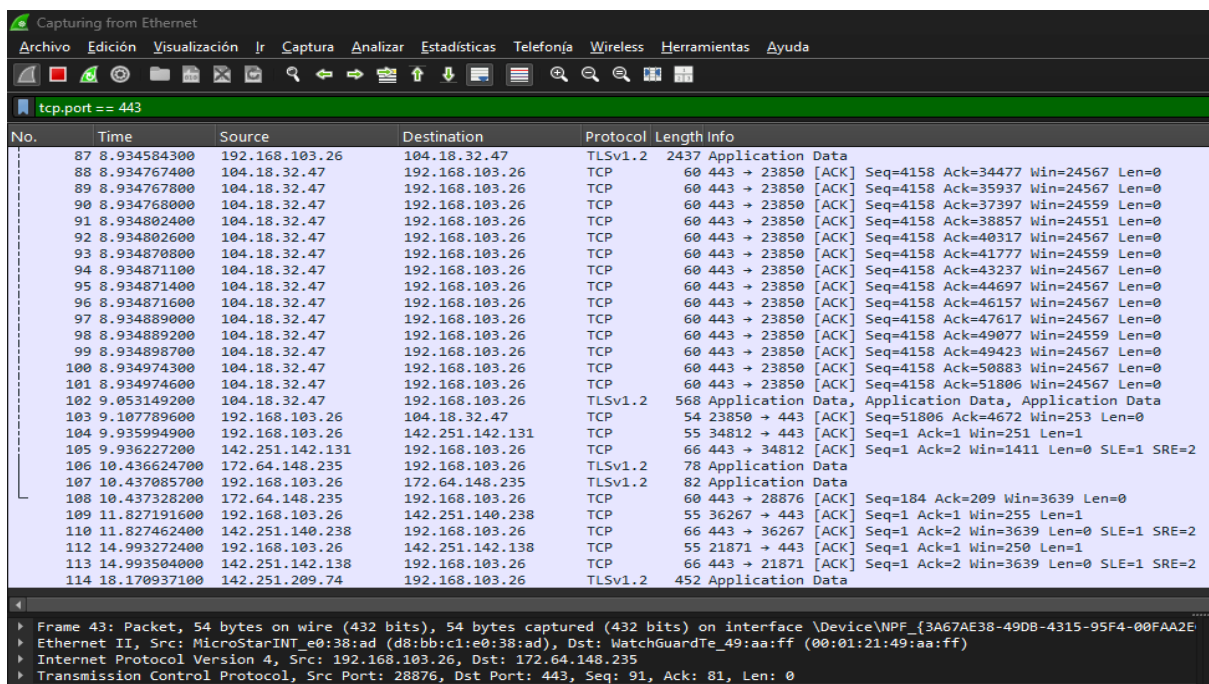
No.	Time	Source	Destination	Protocol	Length	Info
40	1.336684000	52.110.20.5	192.168.103.26	TCP	60	443 → 21809 [FIN, ACK] Seq=1 Ack=2 Win=1405 Len=0
41	1.336695200	52.110.20.5	192.168.103.26	TCP	60	443 → 21807 [FIN, ACK] Seq=1 Ack=2 Win=1405 Len=0
42	1.336750200	192.168.103.26	52.110.20.5	TCP	54	21809 → 443 [ACK] Seq=2 Ack=2 Win=254 Len=0
43	1.336814300	192.168.103.26	52.110.20.5	TCP	54	21807 → 443 [ACK] Seq=2 Ack=2 Win=253 Len=0
44	1.337855200	2.23.237.142	192.168.103.26	TLSv1.2	85	Encrypted Alert
45	1.337857300	2.23.237.142	192.168.103.26	TCP	60	443 → 21810 [FIN, ACK] Seq=32 Ack=2 Win=360 Len=0
46	1.337977400	192.168.103.26	2.23.237.142	TCP	54	21810 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
47	1.338123800	5.255.145.33	192.168.103.26	TLSv1.2	85	Encrypted Alert
48	1.338125100	5.255.145.33	192.168.103.26	TCP	60	443 → 45406 [FIN, ACK] Seq=32 Ack=2 Win=375 Len=0
49	1.338190400	192.168.103.26	5.255.145.33	TCP	54	45406 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
50	1.360601700	172.187.61.142	192.168.103.26	TCP	60	443 → 45405 [FIN, ACK] Seq=1 Ack=2 Win=375 Len=0
51	1.360690900	192.168.103.26	172.187.61.142	TCP	54	45405 → 443 [ACK] Seq=2 Ack=2 Win=253 Len=0
52	1.379947800	192.168.103.26	142.250.154.94	TCP	54	51693 → 443 [ACK] Seq=2092 Ack=40 Win=255 Len=0
53	1.380169900	142.250.154.94	192.168.103.26	TLSv1.2	191	Application Data, Application Data, Application Data
54	1.380502900	192.168.103.26	142.250.154.94	TLSv1.2	93	Application Data
55	1.380518300	192.168.103.26	142.250.154.94	TLSv1.2	89	Application Data
56	1.380862800	142.250.154.94	192.168.103.26	TCP	60	443 → 51693 [ACK] Seq=177 Ack=2166 Win=3639 Len=0
57	1.396483900	192.168.103.26	51.11.192.48	TCP	54	45404 → 443 [FIN, ACK] Seq=1 Ack=1 Win=1024 Len=0
58	1.437449600	51.11.192.48	192.168.103.26	TCP	60	443 → 45404 [FIN, ACK] Seq=1 Ack=2 Win=3639 Len=0
59	1.437510100	192.168.103.26	51.11.192.48	TCP	54	45404 → 443 [ACK] Seq=2 Ack=2 Win=1024 Len=0
60	2.129457800	192.168.103.26	142.251.20.94	TCP	55	18090 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
61	2.129714900	142.251.20.94	192.168.103.26	TCP	66	443 → 18090 [ACK] Seq=1 Ack=2 Win=1411 Len=0 SLE=1 SRE=2
62	2.870508900	192.178.25.110	192.168.103.26	TLSv1.2	127	Application Data
63	2.870849300	192.168.103.26	192.178.25.110	TCP	54	27275 → 443 [FIN, ACK] Seq=1 Ack=74 Win=255 Len=0
64	2.913738000	192.178.25.110	192.168.103.26	TCP	60	443 → 27275 [ACK] Seq=74 Ack=2 Win=3435 Len=0
65	2.927765100	192.178.25.110	192.168.103.26	TCP	60	443 → 27275 [FIN, ACK] Seq=74 Ack=2 Win=3435 Len=0
66	2.927805400	192.168.103.26	192.178.25.110	TCP	54	27275 → 443 [ACK] Seq=2 Ack=75 Win=255 Len=0

Frame 1: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{3A67AE38-49DB-4315-95F4-00FAA2EC} Ethernet II, Src: MicroStarINT_e0:38:ad (d8:bb:c1:e0:38:ad), Dst: WatchGuardTe_49:aa:ff (00:01:21:49:aa:ff)

Internet Protocol Version 4, Src: 192.168.103.26, Dst: 64.233.184.118

Transmission Control Protocol, Src Port: 4363, Dst Port: 443, Seq: 1, Ack: 1, Len: 1

1. Deseamos visualizar todos los tcp que circulen por el puerto 443. **Filtro tcp.port == 443**



Filter:

No.	Time	Source	Destination	Protocol	Length	Info
87	8.934584300	192.168.103.26	104.18.32.47	TLSv1.2	2437	Application Data
88	8.934767400	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=34477 Win=24567 Len=0
89	8.934767800	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=35937 Win=24567 Len=0
90	8.934768000	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=37397 Win=24559 Len=0
91	8.934802400	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=38857 Win=24551 Len=0
92	8.934802600	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=40317 Win=24567 Len=0
93	8.934870800	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=41777 Win=24559 Len=0
94	8.934871100	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=43237 Win=24567 Len=0
95	8.934871400	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=44697 Win=24567 Len=0
96	8.934871600	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=46157 Win=24567 Len=0
97	8.934889000	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=47617 Win=24567 Len=0
98	8.934889200	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=49077 Win=24559 Len=0
99	8.934898700	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=49423 Win=24567 Len=0
100	8.934974300	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=50883 Win=24567 Len=0
101	8.934974600	104.18.32.47	192.168.103.26	TCP	60	443 → 23850 [ACK] Seq=4158 Ack=51806 Win=24567 Len=0
102	9.053149200	104.18.32.47	192.168.103.26	TLSv1.2	568	Application Data, Application Data, Application Data
103	9.107789600	192.168.103.26	104.18.32.47	TCP	54	23850 → 443 [ACK] Seq=51806 Ack=4672 Win=253 Len=0
104	9.935949000	192.168.103.26	142.251.142.131	TCP	55	34812 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
105	9.936227200	142.251.142.131	192.168.103.26	TCP	66	443 → 34812 [ACK] Seq=1 Ack=2 Win=1411 Len=0 SLE=1 SRE=2
106	10.436624700	172.64.148.235	192.168.103.26	TLSv1.2	78	Application Data
107	10.437085700	192.168.103.26	172.64.148.235	TLSv1.2	82	Application Data
108	10.437328200	172.64.148.235	192.168.103.26	TCP	60	443 → 28876 [ACK] Seq=184 Ack=209 Win=3639 Len=0
109	11.827191600	192.168.103.26	142.251.140.238	TCP	55	36267 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
110	11.827462400	142.251.140.238	192.168.103.26	TCP	66	443 → 36267 [ACK] Seq=1 Ack=2 Win=3639 Len=0 SLE=1 SRE=2
112	14.993272400	192.168.103.26	142.251.142.138	TCP	55	21871 → 443 [ACK] Seq=1 Ack=1 Win=250 Len=1
113	14.993504000	142.251.142.138	192.168.103.26	TCP	66	443 → 21871 [ACK] Seq=1 Ack=2 Win=3639 Len=0 SLE=1 SRE=2
114	18.170937100	142.251.209.74	192.168.103.26	TLSv1.2	452	Application Data

Frame 43: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{3A67AE38-49DB-4315-95F4-00FAA2EC} Ethernet II, Src: MicroStarINT_e0:38:ad (d8:bb:c1:e0:38:ad), Dst: WatchGuardTe_49:aa:ff (00:01:21:49:aa:ff)

Internet Protocol Version 4, Src: 192.168.103.26, Dst: 172.64.148.235

Transmission Control Protocol, Src Port: 28876, Dst Port: 443, Seq: 91, Ack: 81, Len: 0

2. Localizar todos los TCP que circulen por el puerto 80. Filtro: tcp.port == 80

Capturing from Ethernet

Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
150	23.029177900	192.168.103.26	142.251.209.67	TCP	54	26579 → 80 [FIN, ACK] Seq=1 Ack=1 Win=255 Len=0
151	23.029241600	192.168.103.26	142.251.209.67	TCP	54	26580 → 80 [FIN, ACK] Seq=1 Ack=1 Win=251 Len=0
152	23.029516900	142.251.209.67	192.168.103.26	TCP	60	80 → 26580 [FIN, ACK] Seq=1 Ack=2 Win=350 Len=0
153	23.029517200	142.251.209.67	192.168.103.26	TCP	60	80 → 26579 [FIN, ACK] Seq=1 Ack=2 Win=350 Len=0
154	23.029553400	192.168.103.26	142.251.209.67	TCP	54	26580 → 80 [ACK] Seq=2 Ack=2 Win=251 Len=0
155	23.029573000	192.168.103.26	142.251.209.67	TCP	54	26579 → 80 [ACK] Seq=2 Ack=2 Win=255 Len=0

Frame 155: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{3A67AE38-49D8-4315-95F4-00FAA2...}

Ethernet II, Src: MicroStarINT-e0:38:ad (d8:bb:c1:e0:38:ad), Dst: MatchGuardTe_49:aa:ff (00:01:21:49:aa:ff)

Internet Protocol Version 4, Src: 192.168.103.26, Dst: 142.251.209.67

Transmission Control Protocol, Src Port: 26579, Dst Port: 80, Seq: 2, Ack: 2, Len: 0

3. Localizar todos los UDP que circulen por el puerto 53. Filtro: udp.port == 53

Capturing from Ethernet

Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

udp.port == 53

No.	Time	Source	Destination	Protocol	Length	Info
4056	203.554111300	192.168.103.26	8.8.8.8	DNS	80	Standard query 0x0001 PTR 8.8.8.8.in-addr.arpa
4057	203.564063300	8.8.8.8	192.168.103.26	DNS	104	Standard query response 0x0001 PTR 8.8.8.8.in-addr.arpa PTR dns.google
4058	203.566725400	192.168.103.26	8.8.8.8	DNS	70	Standard query 0x0002 A google.com
4059	203.568012000	8.8.8.8	192.168.103.26	DNS	86	Standard query response 0x0002 A google.com A 216.58.205.238
4060	203.591220200	192.168.103.26	8.8.8.8	DNS	70	Standard query 0x0003 AAAA google.com
4061	203.624040600	8.8.8.8	192.168.103.26	DNS	98	Standard query response 0x0003 AAAA google.com AAAA 2a00:1450:4003:811:200e

Frame 4056: Packet, 80 bytes on wire (640 bits), 80 bytes captured (640 bits) on interface \Device\NPF_{3A67AE38-49D8-4315-95F4-00FAA2...}

Ethernet II, Src: MicroStarINT-e0:38:ad (d8:bb:c1:e0:38:ad), Dst: MatchGuardTe_49:aa:ff (00:01:21:49:aa:ff)

Internet Protocol Version 4, Src: 192.168.103.26, Dst: 8.8.8.8

User Datagram Protocol, Src Port: 58264, Dst Port: 53

Domain Name System (query)

4. Localizar todos los paquetes que circulen por UDP. Filtro: udp

Capturing from Ethernet

Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

udp

No.	Time	Source	Destination	Protocol	Length	Info
4459	144.768645900	192.168.103.25	239.255.255.250	SSDP	210	Pi-SEARCH * HTTP/1.1
4463	145.789399400	192.168.103.25	239.255.255.250	SSDP	210	Pi-SEARCH * HTTP/1.1
4520	146.770723300	192.168.103.25	239.255.255.250	SSDP	210	Pi-SEARCH * HTTP/1.1
4538	147.772054400	192.168.103.25	239.255.255.250	SSDP	210	Pi-SEARCH * HTTP/1.1
5125	177.247613300	162.159.190.2	192.168.103.226	ISAKMP	88	Unknown 82[Malformed Packet]
5126	177.247721400	162.159.190.2	192.168.103.226	ISAKMP	88	Unknown 82[Malformed Packet]
5127	177.247958700	162.159.190.2	192.168.103.226	ISAKMP	88	Unknown 82
5128	177.248518100	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82[Malformed Packet]
5129	177.248517900	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82[Malformed Packet]
5130	177.248518000	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82[Malformed Packet]
5131	177.248520000	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82[Malformed Packet]
5132	177.248648800	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82[Malformed Packet]
5133	177.248648900	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82[Malformed Packet]
5134	177.248651100	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82[Malformed Packet]
5135	177.248653300	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82
5136	177.248766800	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82[Malformed Packet]
5137	177.248770000	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82[Malformed Packet]
5138	177.248772000	162.159.190.2	192.168.103.226	ISAKMP	1363	Unknown 82[Malformed Packet]
5945	221.740520400	192.168.103.20	192.168.103.255	BROADCAST	240	Host Announcement (DESKTOP-CESR2J3, Workstation, Server, NT Workstation)
6081	226.425430100	192.168.103.26	8.8.8.8	DNS	87	Standard query 0x7961 A roaming.svc.cloud.microsoft
6082	226.454637700	192.168.103.26	1.1.1.1	DNS	87	Standard query 0x7961 A roaming.svc.cloud.microsoft
6083	226.467066000	1.1.1.1	192.168.103.26	DNS	380	Standard query response 0x7961 A roaming.svc.cloud.microsoft CNAMEx prod.roaming1.live.com.akadns.net CNAMEx eur.roaming1.live.com.akadns.net CNAMEx roaming-prod-weightedw.trafficmanager.net CNAMEx
6089	226.487461300	8.8.8.8	192.168.103.26	DNS	380	Standard query response 0x7961 A roaming.svc.cloud.microsoft CNAMEx prod.roaming1.live.com.akadns.net CNAMEx eur.roaming1.live.com.akadns.net CNAMEx roaming-prod-weightedw.trafficmanager.net CNAMEx
6047	254.770671900	192.168.103.25	239.255.255.250	SSDP	210	Pi-SEARCH * HTTP/1.1
6064	265.772034500	192.168.103.25	239.255.255.250	SSDP	210	Pi-SEARCH * HTTP/1.1
6079	266.773051400	192.168.103.25	239.255.255.250	SSDP	210	Pi-SEARCH * HTTP/1.1
7013	267.773629600	192.168.103.25	239.255.255.250	SSDP	210	Pi-SEARCH * HTTP/1.1

Frame 132: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface \Device\NPF_{3A67AE38-49D8-4315-95F4-00FAA2...}

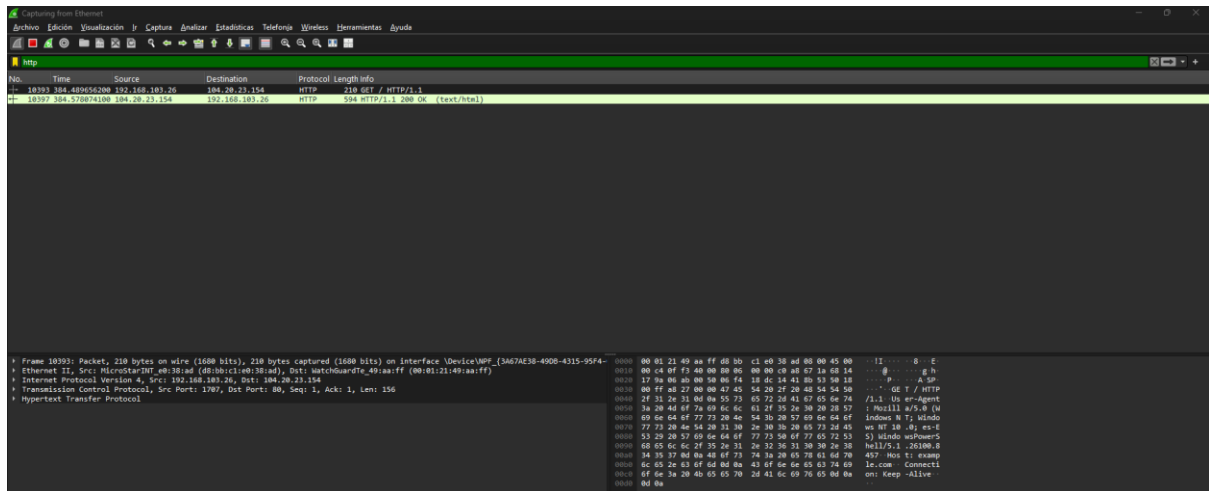
Ethernet II, Src: MicroStarINT-e0:38:ad (d8:bb:c1:e0:38:ad), Dst: MatchGuardTe_49:aa:ff (00:01:21:49:aa:ff)

Internet Protocol Version 4, Src: 192.168.103.26, Dst: 8.8.8.8

User Datagram Protocol, Src Port: 61655, Dst Port: 53

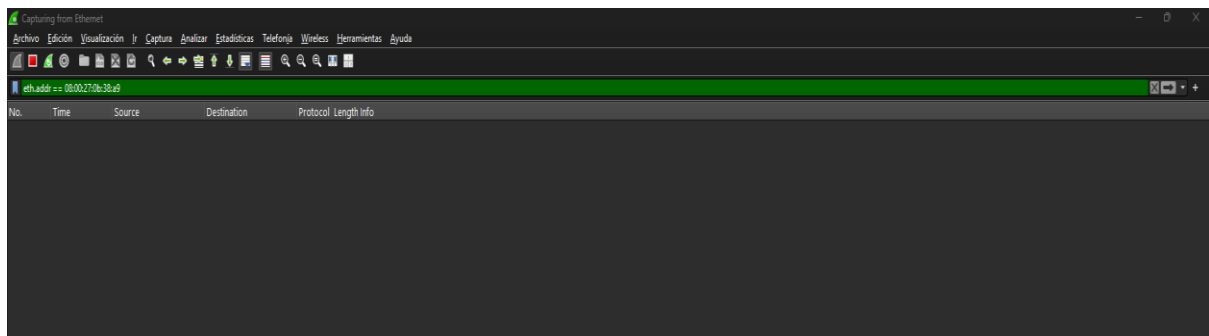
Domain Name System (query)

5. Orden para localizar todos los http. Filtro: http



6. Localizar la MAC 08:00:27:0b:38:a9. Filtro: eth.addr == 08:00:27:0b:38:a9

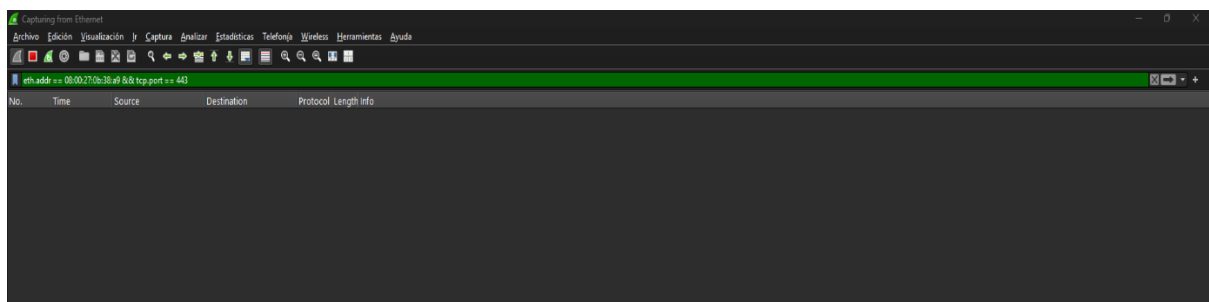
No ha dado resultado debido a que no ha encontrado tráfico asociado a esa dirección MAC en la red



7. Localizar la MAC anterior a través de los tcp “Y” que circulen por el puerto 443.

Filtro: eth.addr == 08:00:27:0b:38:a9 && tcp.port == 443

No ha dado resultado debido a que no ha encontrado tráfico asociado a esa dirección MAC en la red



8. Localizar la ip 192.168.103.26. Filtro: ip.addr == 192.168.103.26

Capturing from eth0

ip.addr == 192.168.103.26

No.	Time	Source	Destination	Protocol	Length	Info
24283	896.732126300	192.168.103.26	216.58.205.238	TLSv1.3	59	Application Data
24284	896.746562000	192.168.103.26	172.217.20.238	TCP	54	50439 → 443 [ACK] Seq=3185599 Ack=1299610 Win=65024 Len=0
24285	896.774712700	216.58.205.238	192.168.103.26	TCP	60	443 → 50439 [ACK] Seq=3185599 Ack=1299610 Win=112832 Len=0
24287	896.982154000	172.217.20.238	192.168.103.26	TLSv1.3	286	Application Data
24288	896.98787700	192.168.103.26	172.217.20.238	TCP	54	50439 → 443 [ACK] Seq=3185599 Ack=1299610 Win=64768 Len=0
24289	896.954672000	172.217.20.238	192.168.103.26	TLSv1.3	259	Application Data
24290	891.802465000	192.168.103.26	172.217.20.238	TCP	54	50439 → 443 [ACK] Seq=3185599 Ack=1300047 Win=64512 Len=0
24291	891.132163000	172.217.20.238	192.168.103.26	TLSv1.3	287	Application Data
24292	891.242295000	192.168.103.26	172.217.20.238	TCP	54	50439 → 443 [ACK] Seq=3185599 Ack=1300208 Win=64256 Len=0
24293	891.695959000	172.217.20.238	192.168.103.26	TLSv1.3	396	Application Data
24294	891.738059000	192.168.103.26	172.217.20.238	TCP	54	50439 → 443 [ACK] Seq=3185599 Ack=1300542 Win=64000 Len=0
24296	892.800061000	172.217.20.238	192.168.103.26	TLSv1.3	248	Application Data
24297	892.040434100	192.168.103.26	172.217.20.238	TCP	54	50439 → 443 [ACK] Seq=3185599 Ack=1300736 Win=65200 Len=0
24298	892.132467500	172.217.20.238	192.168.103.26	TLSv1.3	253	Application Data
24299	892.132855200	192.168.103.26	172.217.20.238	TLSv1.3	89	Application Data
24300	892.132672000	192.168.103.26	172.217.20.238	TLSv1.3	89	Application Data
24301	892.132077000	172.217.20.238	192.168.103.26	TCP	60	443 → 50439 [ACK] Seq=3185599 Ack=1305669 Win=268532 Len=0
24302	892.433994000	172.217.20.238	192.168.103.26	TLSv1.3	224	Application Data
24303	892.454613000	192.168.103.26	172.217.20.238	TCP	54	50439 → 443 [ACK] Seq=3185669 Ack=1301184 Win=65024 Len=0
24304	892.740843000	172.217.20.238	192.168.103.26	TLSv1.3	145	Application Data
24305	892.781493000	172.217.20.238	192.168.103.26	TCP	54	50439 → 443 [ACK] Seq=3185669 Ack=1301195 Win=65024 Len=0
24306	892.797690000	172.217.20.238	192.168.103.26	TLSv1.3	355	Application Data
24307	892.843739000	192.168.103.26	172.217.20.238	TCP	54	50439 → 443 [ACK] Seq=3185669 Ack=1301486 Win=64768 Len=0
24308	891.800774000	172.217.20.238	192.168.103.26	TLSv1.3	338	Application Data
24310	891.123647000	192.168.103.26	172.217.20.238	TCP	54	50439 → 443 [ACK] Seq=3185669 Ack=1301772 Win=64256 Len=0
24311	891.693959000	192.168.103.26	216.58.205.238	TLSv1.3	2943	Application Data

Frame 11 Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on Interface vDeviceWPF_{3A67AE38-4908-4315-95F4-00FA2E2C8C3C}

Ethernet II, Src: MicroStarNet_eb:38:ad (d8bb:bc1e:b3:ad), Dst: WatchGuardTe_9a:aa:ff (00:01:21:49:aa:ff)

Internet Protocol Version 4, Src: 192.168.103.26, Dst: 172.64.155.209

Transmission Control Protocol, Src Port: 4363, Dst Port: 443, Seq: 1, Ack: 1, Len: 1

9. Localizar la ip anterior “O” que circule por el puerto 80.

Filtro: ip.addr == 192.168.103.26 || tcp.port == 80

Capturing from eth0

ip.addr == 192.168.103.26 || tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
38217	1145.1007540	192.168.103.26	172.64.155.209	TLSv1.3	54	Application Data
38218	1145.2320821	172.64.155.209	192.168.103.26	TCP	60	443 → 54326 [ACK] Seq=4407 Ack=11681 Win=405792 Len=0
38219	1145.2354288	172.64.155.209	192.168.103.26	TLSv1.3	139	Application Data
38240	1145.2780334	192.168.103.26	172.64.155.209	TCP	54	54326 → 443 [ACK] Seq=11461 Ack=4802 Win=65200 Len=0
38241	1145.7960593	192.168.103.26	142.251.142.138	TCP	55	[TCP Keep-Alive] 32872 → 443 [ACK] Seq=1342122 Ack=908083 Win=65200 Len=1
38242	1145.7960526	142.251.142.138	192.168.103.26	TCP	66	[TCP Keep-Alive ACK] 443 → 32872 [ACK] Seq=400563 Ack=1342122 Win=2016368 Len=0 SLE=1342122 SRE=1342122
38244	1149.4361388	192.168.103.26	216.58.205.42	TCP	55	[TCP Keep-Alive] 16009 → 443 [ACK] Seq=3674 Ack=3297 Win=65024 Len=1
38245	1149.4363594	216.58.205.42	192.168.103.26	TCP	66	[TCP Keep-Alive ACK] 443 → 3674 [ACK] Seq=3674 Ack=3675 Win=180368 Len=0 SLE=3674 SRE=3675
38247	1152.1664958	192.168.103.26	142.251.150.119	TCP	55	[TCP Keep-Alive] 34105 → 443 [ACK] Seq=3413 Ack=40808 Win=64512 Len=1
38248	1152.1664523	142.251.150.119	192.168.103.26	TCP	66	[TCP Keep-Alive ACK] 443 → 34105 [ACK] Seq=4800 Ack=3414 Win=180608 Len=0 SLE=3413 SRE=3414
38249	1152.2286807	192.168.103.26	142.251.151.119	TCP	55	[TCP Keep-Alive] 34801 → 443 [ACK] Seq=4365 Ack=51945 Win=64256 Len=1
38250	1152.2289222	142.251.151.119	192.168.103.26	TCP	66	[TCP Keep-Alive ACK] 443 → 34801 [ACK] Seq=51945 Ack=4368 Win=117440 Len=0 SLE=4365 SRE=4366
38251	1152.4518081	192.168.103.26	216.58.205.78	TCP	55	[TCP Keep-Alive] 24454 → 443 [ACK] Seq=4858 Ack=4971 Win=64256 Len=1
38252	1152.4520848	216.58.205.78	192.168.103.26	TCP	66	[TCP Keep-Alive ACK] 443 → 24454 [ACK] Seq=4971 Ack=4859 Win=311552 Len=0 SLE=4858 SRE=4859
38253	1152.4094316	192.168.103.26	216.58.205.42	TCP	55	[TCP Keep-Alive] 35842 → 443 [ACK] Seq=5329 Ack=3613 Win=65200 Len=1
38254	1152.4096773	216.58.205.42	192.168.103.26	TCP	66	[TCP Keep-Alive ACK] 443 → 35842 [ACK] Seq=3613 Ack=5338 Win=436800 Len=0 SLE=5329 SRE=5330
38255	1152.8342159	192.168.103.26	172.217.20.225	TCP	55	[TCP Keep-Alive] 13297 → 443 [ACK] Seq=3902 Ack=3369 Win=65200 Len=1
38256	1152.8342426	172.217.20.225	192.168.103.26	TCP	66	[TCP Keep-Alive ACK] 443 → 13297 [ACK] Seq=3369 Ack=3903 Win=380608 Len=0 SLE=3902 SRE=3903
38257	1152.8657971	192.168.103.26	74.125.206.188	TCP	55	[TCP Keep-Alive] 58274 → 443 [ACK] Seq=2009 Ack=6000 Win=64000 Len=1
38258	1152.8661867	74.125.206.188	192.168.103.26	TCP	66	[TCP Keep-Alive ACK] 443 → 58274 [ACK] Seq=4900 Ack=2018 Win=180608 Len=0 SLE=2009 SRE=2010
38259	1151.4740809	192.168.103.26	216.58.205.238	TCP	55	[TCP Keep-Alive] 37314 → 443 [ACK] Seq=9332 Ack=3813 Win=64256 Len=1
38260	1151.4742689	216.58.205.238	192.168.103.26	TCP	66	[TCP Keep-Alive ACK] 443 → 37314 [ACK] Seq=3813 Ack=9333 Win=442624 Len=0 SLE=5532 SRE=5533
38261	1151.7121817	192.168.103.26	216.58.205.238	TCP	55	[TCP Keep-Alive] 11597 → 443 [ACK] Seq=3145593 Ack=130808 Win=64512 Len=1
38262	1151.7124326	216.58.205.238	192.168.103.26	TCP	66	[TCP Keep-Alive ACK] 443 → 11597 [ACK] Seq=31882 Ack=3145594 Win=112832 Len=0 SLE=3145593 SRE=3145594
38263	1154.8007827	172.64.148.235	192.168.103.26	TLSv1.3	78	Application Data
38264	1154.8008087	192.168.103.26	172.64.148.235	TCP	54	50439 → 443 [ACK] Seq=3185669 Ack=1301772 Win=64256 Len=0
38265	1154.8042356	172.64.148.235	192.168.103.26	TCP	60	443 → 54326 [ACK] Seq=4516 Ack=13489 Win=405792 Len=0

Frame 18216 Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on Interface vDeviceWPF_{3A67AE38-4908-4315-95F4-00FA2E2C8C3C}

Section Number: 1

Interface Eth 0 (vDeviceWPF_{3A67AE38-4908-4315-95F4-00FA2E2C8C3C})

Encapsulation type: Ethernet (1)

Arrival Time: May 26, 2026 12:04:15.16618080 Hora de verano romane

UTC Arrival Time: May 26, 2026 10:04:15.16618080 UTC

Epoch Arrival Time: 1779789865.16618080

[Time offset for this packet: 0.00000000 seconds]

[Time delta from previous captured frame: 15.685500 milliseconds]

[Time delta from previous displayed frame: 15.685500 milliseconds]

[Time since reference or first frame: 6 minutes, 15.37063700 seconds]

Frame Number: 18216

Frame Length: 54 bytes (432 bits)

Capture Length: 54 bytes (432 bits)

[Frame is loaded: false]

[Frame is ignored: false]

[Protocols in frame: ethertypeip:tcp]

Character encoding: ASCII (0)

[Coloring Rule Name: TCP]

[Coloring Rule String: tcp]

Ethernet II, Src: MicroStarNet_eb:38:ad (d8bb:bc1e:b3:ad), Dst: WatchGuardTe_9a:aa:ff (00:01:21:49:aa:ff)

Destination: WatchGuardTe_9a:aa:ff (00:01:21:49:aa:ff)

Source: MicroStarNet_eb:38:ad (d8bb:bc1e:b3:ad)

Type: IPv4 (0x0800)

[Stream Index: 8]

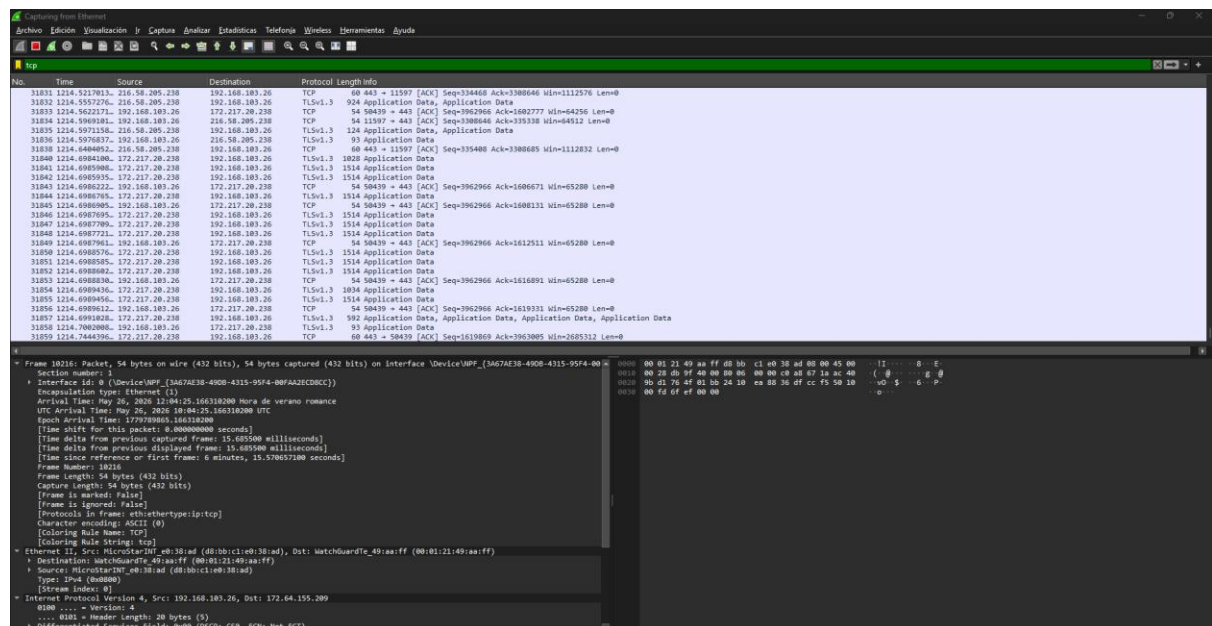
Internet Protocol Version 4, Src: 192.168.103.26, Dst: 172.64.155.209

0x00 = Version: 4

.... 0x00 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

10. Localizar todo el tráfico tcp. Filtro: tcp



Wireshark capture showing TCP traffic. The packet list shows multiple TCP segments. The packet details pane shows the structure of a TCP segment, including the header and application data.

Frame 10216: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface VDeviceUPP (3A67AE38-4908-4315-95F4-00...)

Section numbers:

- Interface 10: VDeviceUPP (3A67AE38-4908-4315-95F4-00FAE2C0C0C)
- Encapsulation type: Ethernet (1)
- Arrival Time: May 26, 2026 10:04:25.166318000 Hora de verano romance
- UTC Arrival Time: May 26, 2026 10:04:25.166318000 UTC
- Ethernet II, Src: MicroStarInt_eb38:ad (d8bb:1c:e0:38:ad), Dst: WatchGuardT_49:aa:ff (00:01:21:49:aa:ff)
- Time delta from previous captured frame: 15.685500 milliseconds
- Time delta from previous displayed frame: 15.685500 milliseconds
- Time since reference or first frame: 6 minutes, 15.570657100 seconds
- Frame Number: 10216
- Frame Length: 54 bytes (432 bits)
- Capture Length: 54 bytes (432 bits)
- Frame is marked: false
- Frame is ignored: false
- Protocol in frame: ethertype:ip:tcp
- Character encoding: ASCII (0)
- Coloring Rule Name: TCP
- Coloring Rule String: tcp

Ethernet II, Src: MicroStarInt_eb38:ad (d8bb:1c:e0:38:ad), Dst: WatchGuardT_49:aa:ff (00:01:21:49:aa:ff)

- Destination: WatchGuardT_49:aa:ff (00:01:21:49:aa:ff)
- Source: MicroStarInt_eb38:ad (d8bb:1c:e0:38:ad)
- Type: TCP (6)
- Stream index: 0

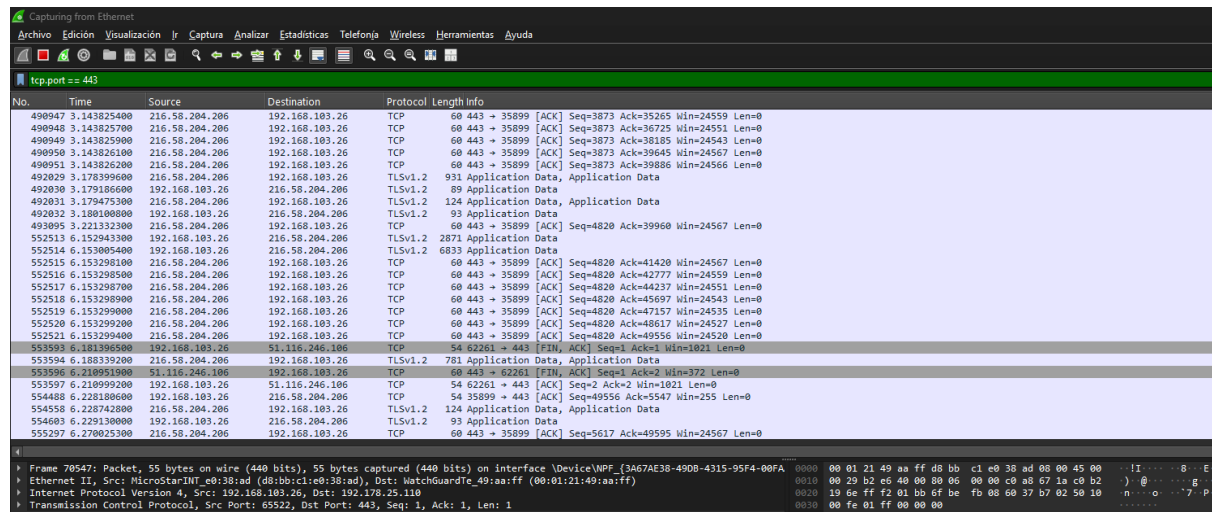
Internet Protocol Version 4, Src: 192.168.103.26, Dst: 172.64.155.209

6500 ... = Version: 4

.... 0101 = Header Length: 20 bytes (5)

= Differentiated Services Field: 0x00 (CS0: CS0, ECN: Not-ECT)

11. Localizar todo el tráfico HTTPS. Filtro: tcp.port == 443



Wireshark capture showing HTTPS traffic. The packet list shows multiple TCP segments. The packet details pane shows the structure of a TCP segment, including the header and application data.

Frame 70547: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface VDeviceNPF (3A67AE38-4908-4315-95F4-00FAE2C0C0C)

Ethernet II, Src: MicroStarInt_eb38:ad (d8bb:1c:e0:38:ad), Dst: WatchGuardT_49:aa:ff (00:01:21:49:aa:ff)

- Destination: WatchGuardT_49:aa:ff (00:01:21:49:aa:ff)
- Source: MicroStarInt_eb38:ad (d8bb:1c:e0:38:ad)
- Type: Transmission Control Protocol, Src Port: 65522, Dst Port: 443, Seq: 1, Ack: 1, Len: 1

12. Localizar todo el tráfico HTTP. Filtro: tcp.port == 80

Wireshark capture showing HTTP traffic filtered by tcp.port == 80. The packet list shows multiple TCP segments from 192.168.103.26 to 192.168.103.26. The packet details pane shows the structure of a TCP segment, including the header and payload.

13. Mostrar únicamente tráfico DNS por UDP. Filtro: udp.port == 53

Wireshark capture showing DNS traffic filtered by udp.port == 53. The packet list shows several DNS queries and responses. The packet details pane shows the structure of a DNS query, including the header and the question section.

14. Buscar todos los paquetes ICMP (ping). Filtro: icmp

Wireshark capture showing ICMP traffic filtered by icmp. The packet list shows several ICMP echo request and echo reply packets. The packet details pane shows the structure of an ICMP echo request, including the header and the data field.

15. Localizar tráfico de la IP 192.168.103.26. Filtro: ip.addr == 192.168.103.26

Capturing from Ethernet

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ip.addr == 192.168.103.26

No.	Time	Source	Destination	Protocol	Length	Info
519	52.066131900	216.58.205.42	192.168.103.26	TCP	60	443 → 54926 [ACK] Seq=2739 Ack=10611 Win=13301 Len=0
520	52.066132100	216.58.205.42	192.168.103.26	TCP	60	443 → 54926 [ACK] Seq=2739 Ack=10916 Win=13301 Len=0
521	52.112074100	52.168.112.67	192.168.103.26	TLSv1.2	92	Application Data
522	52.112311200	192.168.103.26	142.251.142.142	TLSv1.2	3951	Application Data
523	52.112353200	192.168.103.26	142.251.142.142	TLSv1.2	2074	Application Data
524	52.112723100	142.251.142.142	192.168.103.26	TCP	60	443 → 62700 [ACK] Seq=13909 Ack=66202 Win=22515 Len=0
525	52.112723600	142.251.142.142	192.168.103.26	TCP	60	443 → 62700 [ACK] Seq=13909 Ack=68719 Win=22515 Len=0
526	52.115133300	216.58.205.42	192.168.103.26	TLSv1.2	1344	Application Data, Application Data
527	52.154607100	142.251.142.142	192.168.103.26	TCP	60	443 → 62700 [ACK] Seq=13909 Ack=69279 Win=22515 Len=0
528	52.157720700	192.168.103.26	52.168.112.67	TCP	54	52874 → 443 [ACK] Seq=18666 Ack=4394 Win=65280 Len=0
529	52.157753600	192.168.103.26	216.58.205.42	TCP	54	54926 → 443 [ACK] Seq=10916 Ack=4029 Win=255 Len=0
530	52.157928700	216.58.205.42	192.168.103.26	TLSv1.2	124	Application Data, Application Data
531	52.158961500	192.168.103.26	216.58.205.42	TLSv1.2	93	Application Data
532	52.180269600	216.58.205.42	192.168.103.26	TCP	60	443 → 54926 [ACK] Seq=4099 Ack=10955 Win=13301 Len=0
533	52.283307900	52.168.112.67	192.168.103.26	TLSv1.2	342	Application Data
534	52.286599600	192.168.103.26	52.168.112.67	TCP	54	52874 → 443 [FIN, ACK] Seq=18666 Ack=4682 Win=65024 Len=0
535	52.330621900	52.168.112.67	192.168.103.26	TCP	60	443 → 52874 [ACK] Seq=4682 Ack=18667 Win=465792 Len=0
536	52.334274800	142.251.142.142	192.168.103.26	TLSv1.2	215	Application Data
537	52.376067200	192.168.103.26	142.251.142.142	TCP	54	62700 → 443 [ACK] Seq=69279 Ack=14070 Win=2044 Len=0
538	52.376307600	142.251.142.142	192.168.103.26	TLSv1.2	1367	Application Data, Application Data, Application Data
539	52.380767500	192.168.103.26	142.251.142.142	TLSv1.2	93	Application Data
540	52.381165700	142.251.142.142	192.168.103.26	TCP	60	443 → 62700 [ACK] Seq=15383 Ack=69318 Win=22515 Len=0
541	52.415128200	52.168.112.67	192.168.103.26	TLSv1.2	96	Application Data
542	52.415197300	192.168.103.26	52.168.112.67	TCP	54	52874 → 443 [RST, ACK] Seq=18667 Ack=4724 Win=0 Len=0
543	52.440903900	142.251.142.142	192.168.103.26	TLSv1.2	234	Application Data
544	52.692802600	192.168.103.26	142.251.142.142	TCP	54	62700 → 443 [ACK] Seq=69318 Ack=15563 Win=2047 Len=0
547	54.392917800	192.168.103.26	216.58.204.206	TLSv1.2	2868	Application Data

4

Frame 283: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{3A67AE38-4908-4315-95F4-00FAA2} (d8:bb:c1:e0:38:ad)

Ethernet II, Src: WatchGuardTe_49:aa:ff (08:01:21:49:aa:ff), Dst: MicroStarINT_e0:38:ad (d8:bb:c1:e0:38:ad)

Internet Protocol Version 4, Src: 8.8.4.4, Dst: 192.168.103.26

Transmission Control Protocol, Src Port: 443, Dst Port: 19163, Seq: 1, Ack: 2, Len: 0

0000 d8 bb c1 e0 38 ad 00 01 21 49 aa ff 08 00 45 00 ... 8 ... I ... E

0010 00 34 47 53 40 00 40 06 bf a2 08 08 04 04 c0 a0 ... 465 @ ...

0020 67 1a 01 00 4a d0 87 98 a5 8e 83 ab c3 09 08 10 ... 7 ...

0030 00 f3 ee 1f 00 00 01 01 05 0e 83 ab c3 08 83 ab ...

0040 c3 09

16. Mostrar únicamente tráfico origen desde la IP 192.168.103.30

Filtro: ip.src == 192.168.103.30

Capturing from Ethernet

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ip.src == 192.168.103.30

No.	Time	Source	Destination	Protocol	Length	Info
-----	------	--------	-------------	----------	--------	------

17. Mostrar únicamente tráfico destino hacia la IP 192.168.103.28

Filtro: ip.dst == 192.168.103.28

Capturing from Ethernet

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ip.dst == 192.168.103.28

No.	Time	Source	Destination	Protocol	Length	Info
-----	------	--------	-------------	----------	--------	------

18. Buscar tráfico TCP y HTTPS al mismo tiempo. Filtro: tcp && tcp.port == 443

Wireshark interface showing a packet capture filter: `tcp && tcp.port == 443`. The packet list shows several TCP packets, including a SYN packet (Seq=58166, Ack=1, Win=3639, Len=1460) and an ACK packet (Seq=1514, Ack=1, Win=1365, Len=0, SLE=4294965837, SRE=1).

No.	Time	Source	Destination	Protocol	Length	Info
87	5.069777000	17.250.90.4	192.168.103.24	TCP	1514	443 → 58166 [ACK] Seq=1 Ack=1 Win=3639 Len=1460
163	7.375516000	17.253.122.196	192.168.103.24	TCP	66	443 → 58175 [SYN, ACK, ECE] Seq=0 Ack=1 Win=43690 Len=0 MSS=1460 SACK_PERM WS=128
705	16.014980000	104.18.23.222	192.168.102.119	TLSv1.2	245	Application Data, Application Data
718	16.019782000	104.18.23.222	192.168.102.119	TLSv1.2	1514	Application Data, Application Data
719	16.019784000	104.18.23.222	192.168.102.119	TCP	1514	443 → 51776 [ACK] Seq=1652 Ack=1 Win=1844 Len=1460 [TCP PDU reassembled in 724]
720	16.019790000	104.18.23.222	192.168.102.119	TCP	1514	443 → 51776 [ACK] Seq=3112 Ack=1 Win=1844 Len=1460 [TCP PDU reassembled in 724]
723	16.019958100	104.18.23.222	192.168.102.119	TCP	1514	443 → 51776 [ACK] Seq=4572 Ack=1 Win=1844 Len=1460 [TCP PDU reassembled in 724]
724	16.019965200	104.18.23.222	192.168.102.119	TLSv1.2	1514	Application Data
900	16.540093000	84.16.253.117	192.168.101.61	TLSv1.2	86	Application Data
1494	46.066432200	104.18.29.234	192.168.100.246	TLSv1.2	140	Application Data
2404	91.129182300	17.250.90.4	192.168.101.78	TCP	66	443 → 54711 [ACK] Seq=1 Ack=1 Win=1365 Len=0 SLE=4294965837 SRE=1

19. Localizar todos los paquetes que no pertenezcan a la IP 192.168.103.24.

Filtro: not ip.addr == 192.168.103.24

Wireshark interface showing a packet capture filter: `not ip.addr == 192.168.103.24`. The packet list shows several packets, including ARP requests and responses, and various application data packets.

No.	Time	Source	Destination	Protocol	Length	Info		
2878	115.767330900	185.132.178.104	192.168.103.26	ISAOKP	154	[Malformed Packet]		
2879	115.791208900	185.132.178.104	192.168.103.26	ISAOKP	234			
2880	115.792165700	185.132.178.104	192.168.103.26	ISAOKP	138	[Malformed Packet]		
2881	115.792921600	192.168.103.26	185.132.178.104	ISAOKP	138	[Malformed Packet]		
2882	115.796339100	185.132.178.104	192.168.103.26	ISAOKP	666			
2883	115.796342000	185.132.178.104	192.168.103.26	ISAOKP	138			
2884	115.796776000	192.168.103.26	185.132.178.104	ISAOKP	138	[Malformed Packet]		
2885	115.823333400	192.168.103.26	185.132.178.104	ISAOKP	122	[Malformed Packet]		
2886	115.828719500	185.132.178.104	192.168.103.26	ISAOKP	1466			
2887	115.829214500	192.168.103.26	185.132.178.104	ISAOKP	154			
2888	115.863003400	185.132.178.104	192.168.103.26	ISAOKP	810			
2889	115.863823100	192.168.103.26	185.132.178.104	ISAOKP	154			
2890	115.867343900	185.132.178.104	192.168.103.26	ISAOKP	138	[Malformed Packet]		
2891	115.886798900	MicroStarINT_e0:38::: WatchGuardTe_49:aa::ff	42 who has 192.168.100.1? Tell 192.168.103.26	ARP	60	192.168.100.1 is at 00:01:21:49:aa:ff		
2892	115.887068000	WatchGuardTe_49:aa::ff	MicroStarINT_e0:38::: ARP	60	192.168.103.26	ISAOKP	122	
2893	115.903393600	185.132.178.104	192.168.103.26	ISAOKP	138	[Malformed Packet]		
2894	116.432693100	185.132.178.104	192.168.103.26	ISAOKP	122			
2895	116.432696200	185.132.178.104	192.168.103.26	ISAOKP	122			
2896	116.433236000	192.168.103.26	185.132.178.104	ISAOKP	122	[Malformed Packet]		
2897	116.433759700	192.168.103.26	185.132.178.104	ISAOKP	122			
2898	116.472506300	185.132.178.104	192.168.103.26	ISAOKP	122	[Malformed Packet]		
2899	117.214740100	192.168.103.26	185.132.178.104	ISAOKP	122	[Malformed Packet]		
2900	117.253957200	185.132.178.104	192.168.103.26	ISAOKP	138			
2901	118.500549900	OLinkInterna_51:96::: Ethernet-Configurat_... LOOP	60	Unknown function (256)				
2902	118.854329400	185.132.178.104	192.168.103.26	ISAOKP	138	[Malformed Packet]		
2903	118.855245000	192.168.103.26	185.132.178.104	ISAOKP	154	[Malformed Packet]		
2904	118.894309500	185.132.178.104	192.168.103.26	ISAOKP	122	[Malformed Packet]		

20. Buscar paquetes que contengan la palabra “google”. Filtro: frame contains "google"

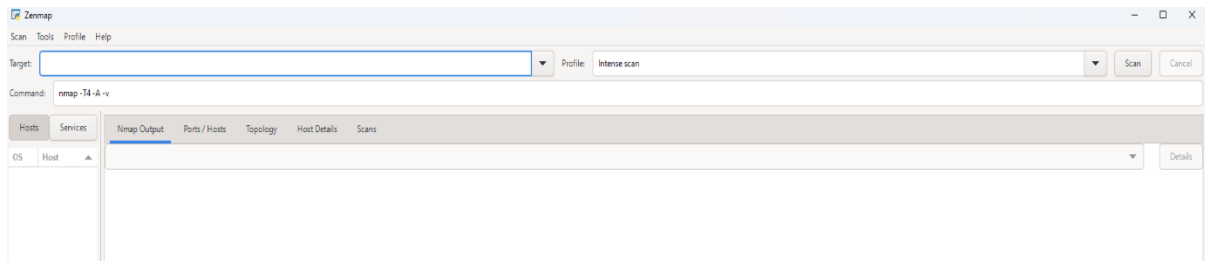
Wireshark interface showing a packet capture filter: `frame contains "google"`. The packet list shows several packets, including a SYN packet (Seq=58166, Ack=1, Win=3639, Len=1460) and an ACK packet (Seq=1514, Ack=1, Win=1365, Len=0, SLE=4294965837, SRE=1).

No.	Time	Source	Destination	Protocol	Length	Info
87	5.069777000	17.250.90.4	192.168.103.24	TCP	1514	443 → 58166 [ACK] Seq=1 Ack=1 Win=3639 Len=1460
163	7.375516000	17.253.122.196	192.168.103.24	TCP	66	443 → 58175 [SYN, ACK, ECE] Seq=0 Ack=1 Win=43690 Len=0 MSS=1460 SACK_PERM WS=128
705	16.014980000	104.18.23.222	192.168.102.119	TLSv1.2	245	Application Data, Application Data
718	16.019782000	104.18.23.222	192.168.102.119	TLSv1.2	1514	Application Data, Application Data
719	16.019784000	104.18.23.222	192.168.102.119	TCP	1514	443 → 51776 [ACK] Seq=1652 Ack=1 Win=1844 Len=1460 [TCP PDU reassembled in 724]
720	16.019790000	104.18.23.222	192.168.102.119	TCP	1514	443 → 51776 [ACK] Seq=3112 Ack=1 Win=1844 Len=1460 [TCP PDU reassembled in 724]
723	16.019958100	104.18.23.222	192.168.102.119	TCP	1514	443 → 51776 [ACK] Seq=4572 Ack=1 Win=1844 Len=1460 [TCP PDU reassembled in 724]
724	16.019965200	104.18.23.222	192.168.102.119	TLSv1.2	1514	Application Data
900	16.540093000	84.16.253.117	192.168.101.61	TLSv1.2	86	Application Data
1494	46.066432200	104.18.29.234	192.168.100.246	TLSv1.2	140	Application Data
2404	91.129182300	17.250.90.4	192.168.101.78	TCP	66	443 → 54711 [ACK] Seq=1 Ack=1 Win=1365 Len=0 SLE=4294965837 SRE=1

NMAP

Nmap es una herramienta de escaneo y análisis de redes utilizada para detectar dispositivos conectados, puertos abiertos y servicios activos dentro de una red. Permite realizar distintos tipos de análisis, desde escaneos rápidos hasta detección de sistemas operativos y versiones de servicios, siendo una herramienta fundamental para administración y seguridad informática.

Interfaz gráfica de NMAP



Se ejecutarán los comandos desde CMD, utilizando Nmap escribiendo directamente “nmap” seguido de las opciones correspondientes en cada instrucción para realizar los distintos escaneos de red.

1. Deseamos escanear de forma silenciosa nuestra red.

Filtro: `nmap -sS 192.168.103.22-33`

```
C:\Program Files (x86)\Nmap>nmap -sS 192.168.103.22-33
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-26 14:20 +0200
Nmap scan report for 192.168.103.22
Host is up (0.00085s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp   open  rpcbind
3128/tcp  open  squid-http
8000/tcp  open  http-alt
9000/tcp  open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.23
Host is up (0.0027s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
2179/tcp  open  vmrpd
MAC Address: 04:7C:16:00:B7:3C (Micro-Star Intl)

Nmap scan report for 192.168.103.25
Host is up (0.00094s latency).
Not shown: 983 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
3000/tcp  open  ppp
8000/tcp  open  http-alt
8081/tcp  open  blackice-icecap
8082/tcp  open  blackice-alerts
8083/tcp  open  us-srv
8084/tcp  open  websnp
8085/tcp  open  unknown
8086/tcp  open  d-s-n
8087/tcp  open  simplifymedia
8088/tcp  open  radan-http
8089/tcp  open  unknown
8099/tcp  open  unknown
9090/tcp  open  zeus-admin
9100/tcp  open  jetdirect
```

2. Queremos conocer la dirección IP de Adams.es. Filtro: `nmap adams.es`

```
C:\Program Files (x86)\Nmap>nmap adams.es
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-26 14:23 +0200
Nmap scan report for adams.es (172.67.71.32)
Host is up (0.0018s latency).
Other addresses for adams.es (not scanned): 2606:4700:20::681a:1d0 2606:4700:20::ac43:4720 2606:4700:20::681a:d0 104.
26.1.208 104.26.0.208
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
8080/tcp  open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 13.49 seconds
```

3. Información del sistema operativo de la máquina de destino.

Filtro: nmap -O 192.168.103.25

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -O 192.168.103.25
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-26 14:24 +0200
Nmap scan report for 192.168.103.25
Host is up (0.00057s latency).
Not shown: 983 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
3000/tcp  open  ppp
8000/tcp  open  http-alt
8081/tcp  open  blackice-icecap
8082/tcp  open  blackice-alerts
8083/tcp  open  us-srv
8084/tcp  open  websnp
8085/tcp  open  unknown
8086/tcp  open  d-s-n
8087/tcp  open  simplifymedia
8088/tcp  open  radan-http
8089/tcp  open  unknown
8099/tcp  open  unknown
9090/tcp  open  zeus-admin
9100/tcp  open  jetdirect
MAC Address: 2C:CF:67:7F:E1:B6 (Raspberry Pi (Trading))
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=5/26%OT=53%CT=1%CU=374777%PV=Y%DS=1%DC=D%G=Y%M=2CCF67%T
OS:M=6A159102P=1686-pc-windows-windows)SEQ(SP=104%GCD=1%ISR=107%TI=Z%CI=Z%
OS:II=I%TS=21)SEQ(SP=104%GCD=1%ISR=10A%TI=Z%CI=Z%II=I%TS=21)SEQ(SP=104%GCD=
OS:1%ISR=10D%TI=Z%CI=Z%II=I%TS=22)SEQ(SP=105%GCD=1%ISR=10A%TI=Z%CI=Z%II=I%T
OS:S=21)SEQ(SP=FF%GCD=1%ISR=10E%TI=Z%CI=Z%II=I%TS=21)OPS(O1=M5B4ST11NW9%O2=
OS:M5B4ST11NW9%O3=M5B4NNT11NW9%O4=M5B4ST11NW9%O5=M5B4ST11NW9%O6=M5B4ST11)WI
OS:N(W1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=3F%W=FA
OS:F0%O=M5B4NNSNW9%CC=Y%Q=)T1(R=Y%DF=Y%T=3F%S=O%A=S+F=AS%RD=0%Q=)T2(R=N)T3
OS:(R=N)T4(R=Y%DF=Y%T=3F%W=0%S=A%A=Z%F=R%O=0%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%
OS:Z%A=S+F=AR%O=0%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=0%RD=0%Q=)T7(R=
OS:Y%DF=Y%T=40%W=0%S=Z%A=S+F=AR%O=0%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%R
OS:IPL=6%RID=6%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.38 seconds
```

4. Escaneo rápido de IP de destino. Filtro: nmap -F 192.168.103.27

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -F 192.168.103.27
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-26 14:25 +0200
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.61 seconds
```


5. Escaneo rápido de IP de destino parecido a nmap T4. Filtro: nmap -T4 192.168.103.30

```
C:\Program Files (x86)\Nmap>nmap -T4 192.168.103.30
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:20 +0200
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.62 seconds
```

6. Necesitamos escanear a la vez varias IP

Filtro: nmap 192.168.103.22 192.168.103.25 192.168.103.31

```
C:\Program Files (x86)\Nmap>nmap 192.168.103.22 192.168.103.25 192.168.103.31
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:21 +0200
Nmap scan report for 192.168.103.22
Host is up (0.00055s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp   open  rpcbind
128/tcp   open  squid-http
8000/tcp  open  http-alt
8000/tcp  open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)
Nmap done: 3 IP addresses (1 host up) scanned in 2.17 seconds
```

7. Información detallada de una IP en concreto según la va encontrando nmap.

Filtro: nmap -sV 192.168.103.24

```
C:\Program Files (x86)\Nmap>nmap -sV 192.168.103.24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:22 +0200
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.85 seconds
```

8. Información de todos los comandos de nmap. Filtro: nmap --help

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap --help
Nmap 7.99 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
```

9. Escanear solo máquinas conectadas (sin puertos). Filtro: nmap -sn 192.168.103.22-33

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -sn 192.168.103.22-33
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:25 +0200
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 11 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 68.18% done; ETC: 09:25 (0:00:00 remaining)
Nmap scan report for 192.168.103.22
Host is up (0.0010s latency).
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)
Nmap scan report for 192.168.103.23
Host is up (0.0010s latency).
MAC Address: 04:7C:16:00:B7:3C (Micro-Star Intl)
Nmap scan report for 192.168.103.27
Host is up (0.0010s latency).
MAC Address: E8:D8:D1:D2:F5:3A (HP)
Nmap scan report for 192.168.103.28
Host is up (0.0010s latency).
MAC Address: 04:7C:16:00:B7:70 (Micro-Star Intl)
Nmap scan report for 192.168.103.29
Host is up (0.0010s latency).
MAC Address: 04:7C:16:00:B7:3B (Micro-Star Intl)
Nmap scan report for 192.168.103.33
Host is up (0.00s latency).
MAC Address: B8:97:5A:6F:CD:F3 (Biostar Microtech Int'l)
Nmap scan report for 192.168.103.26
Host is up.
Nmap done: 12 IP addresses (7 hosts up) scanned in 3.43 seconds
```

10. Escanear los primeros 500 puertos de una IP en concreto.

Filtro: `nmap -p 1-500 192.168.103.28`

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -p 1-500 192.168.103.28
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:26 +0200
Nmap scan report for 192.168.103.28
Host is up (0.0015s latency).
Not shown: 497 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 04:7C:16:00:B7:70 (Micro-Star Intl)

Nmap done: 1 IP address (1 host up) scanned in 1.84 seconds
```

11. Informarse de la IP de Google.es. Filtro: `nmap google.es`

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap google.es
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:27 +0200
Nmap scan report for google.es (172.217.171.35)
Host is up (0.0023s latency).
Other addresses for google.es (not scanned): 2a00:1450:4003:811::2003
DNS record for 172.217.171.35: lmadb-ac-in-f3.1e100.net
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp    open  https

Nmap done: 1 IP address (1 host up) scanned in 9.05 seconds
```

12. Conocer los primeros 800 puertos de Google.es. Filtro: `nmap -p 1-800 google.es`

```
C:\Program Files (x86)\Nmap>nmap -p 1-800 google.es
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:27 +0200
Nmap scan report for google.es (172.217.171.35)
Host is up (0.0021s latency).
Other addresses for google.es (not scanned): 2a00:1450:4003:804::2003
rDNS record for 172.217.171.35: lcmdb-ac-in-f3.1e100.net
Not shown: 797 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 12.97 seconds
```

13. Escanear toda la red. Filtro: `nmap 192.168.103.0/24`

```
C:\Program Files (x86)\Nmap>nmap 192.168.103.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:28 +0200
Nmap scan report for 192.168.103.22
Host is up (0.0015s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp    open  rpcbind
3128/tcp   open  squid-http
8000/tcp   open  http-alt
9000/tcp   open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.23
Host is up (0.0077s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
2179/tcp   open  vmrpd
MAC Address: 04:7C:16:00:B7:3C (Micro-Star Intl)

Nmap scan report for 192.168.103.27
Host is up (0.0029s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
902/tcp    open  iss-realsecure
912/tcp    open  apex-mesh
```

14. Analizar varias máquinas (último octeto). Filtro: nmap 192.168.103.24,26,33

```
C:\Program Files (x86)\Nmap>nmap 192.168.103.24,26,33
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:29 +0200
Nmap scan report for 192.168.103.33
Host is up (0.0028s latency).
Not shown: 982 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
53/tcp    open  domain
80/tcp    open  http
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
143/tcp   open  imap
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
587/tcp   open  submission
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
3389/tcp  open  ms-wbt-server
5357/tcp  open  wsapi
5985/tcp  open  wsman
MAC Address: B8:97:5A:6F:CD:F3 (Biostar Microtech Int'l)

Nmap scan report for 192.168.103.26
Host is up (0.000088s latency).
```

15. Escanear el rango de direcciones. Filtro: nmap 192.168.103.20-50

```
C:\Program Files (x86)\Nmap>nmap 192.168.103.20-50
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:30 +0200
Nmap scan report for 192.168.103.22
Host is up (0.0015s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp   open  rpcbind
3128/tcp  open  squid-http
8000/tcp  open  http-alt
9000/tcp  open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.27
Host is up (0.015s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
445/tcp   open  microsoft-ds
9002/tcp  open  iss-realsecure
912/tcp   open  apex-mesh
2179/tcp  open  vmrdp
3389/tcp  open  ms-wbt-server
5357/tcp  open  wsapi
MAC Address: E8:D8:D1:D2:F5:3A (HP)

Nmap scan report for 192.168.103.28
Host is up (0.0018s latency).
Not shown: 994 closed tcp ports (reset)
```

16. Escanear la red excluyendo una IP.

Filtro: nmap 192.168.103.0/24 --exclude 192.168.103.25

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap 192.168.103.0/24 --exclude 192.168.103.25
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:31 +0200
Nmap scan report for 192.168.103.22
Host is up (0.00073s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp   open  rpcbind
3128/tcp  open  squid-http
8000/tcp  open  http-alt
9000/tcp  open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.23
Host is up (0.0011s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
2179/tcp  open  vmrpd
MAC Address: 04:7C:16:00:B7:3C (Micro-Star Intl)

Nmap scan report for 192.168.103.27
Host is up (0.0011s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
445/tcp   open  microsoft-ds
902/tcp   open  iss-realsecure
```

17. Escanear la red excluyendo un rango de IPs.

Filtro: `nmap 192.168.103.0/24 --exclude 192.168.103.24-28`

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap 192.168.103.0/24 --exclude 192.168.103.24-28
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:31 +0200
Nmap scan report for 192.168.103.22
Host is up (0.0018s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp   open  rpcbind
8128/tcp  open  squid-http
3000/tcp  open  http-alt
9000/tcp  open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.29
Host is up (0.00089s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
9002/tcp  open  iss-realsecure
912/tcp   open  apex-mesh
3306/tcp  open  mysql
5357/tcp  open  wsddapi
MAC Address: 04:7C:16:00:B7:3B (Micro-Star Intl)

Nmap scan report for 192.168.103.30
Host is up (0.00076s latency).
```


18. Qué hace la orden nmap 192.168.103.*. Filtro: escanea toda la red 192.168.103.0/24

```
C:\Program Files (x86)\Nmap>nmap 192.168.103.*
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:32 +0200
Nmap scan report for 192.168.103.22
Host is up (0.00087s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp   open  rpcbind
3128/tcp  open  squid-http
8000/tcp  open  http-alt
9000/tcp  open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.23
Host is up (0.0023s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
2179/tcp  open  vmrpd
MAC Address: 04:7C:16:00:B7:3C (Micro-Star Intl)

Nmap scan report for 192.168.103.27
Host is up (0.0015s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
445/tcp   open  microsoft-ds
9002/tcp  open  iss-realsecure
```

19. Conocer la versión de nmap. Filtro: nmap -V

```
C:\Program Files (x86)\Nmap>nmap -v
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:33 +0200
Read data files from: C:\Program Files (x86)\Nmap
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.05 seconds
Raw packets sent: 0 (0B) | Rcvd: 0 (0B)
```

20. Servidores de prueba Nmap. Filtro: scanme.nmap.org y testphp.nmap.org

```
C:\Program Files (x86)\Nmap>nmap scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:34 +0200
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.16s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
8080/tcp   closed http-proxy

Nmap done: 1 IP address (1 host up) scanned in 28.45 seconds
```

```
C:\Program Files (x86)\Nmap>nmap testphp.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:35 +0200
Nmap scan report for testphp.nmap.org (50.116.1.184)
Host is up (0.0013s latency).
Other addresses for testphp.nmap.org (not scanned): 2600:3c01:e000:3e6::6d4e:7061
rDNS record for 50.116.1.184: ack.nmap.org
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 14.92 seconds
```

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